Scoped Enumeration Stringification and Annotation-Driven Defaultable Formatting with C++26 Static Reflection

Extensions for `std::to_string` and `std::formatter`

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1 Abstract

A short 1–2 paragraph overview explaining the proposal, the problem it solves, and the basic approach being proposed for C++.

2 Revision History

2.1 R0

— Initial revision.

3 Motivation and Scope

Explain why this proposal exists, what real-world use cases it addresses, and the limitations of current C++.

3.1 Before / After Comparison

Before (Current C++)	After (With Proposal)

`// Motivating example showing current verbose, error-prone, or inefficient code`
`#include <iostream>`
 	
`void legacy_way() {`
`  // Current approach`
`}`
	`// Clean code using your proposed feature`
	`void proposed_way() {`
	`  // Proposed syntax or library feature`
	`}`

`void legacy_way() {`
`  // Current approach`
`}`
	`void proposed_way() {`
	`  // Proposed syntax or library feature`
	`}`

4 Design Decisions

Explain the technical design choices, trade-offs, ABI implications, and feature interactions.

4.1 Alternatives Considered

— **Option A:** Description of an alternative approach and reasons for rejection.
— **Option B:** Another candidate syntax or API and its drawbacks.

5 Proposed Wording

Draft edits relative to the C++ Working Draft. Use markup tags like `<ins>added text</ins>` and `<del>removed text</del>`.

Modify Section [general.description] as follows:

The behavior of the function shall be modified to support the new functionality throw an exception.

6 Implementation Experience

Note whether a prototype exists, compiler branches where it has been implemented, or user experience gained from library implementations.

7 References